Yanjun Chen

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Research Interests

Making the environment trainable, the way models are: measuring what each piece (reward models, verifiers, curricula) contributes to the model it trains, across reinforcement learning, large language models, and embodied AI.

Education

The Hong Kong Polytechnic University, Hong Kong — Sept 2024 to present

PhD, Department of Computing. Candidature confirmed June 2026; thesis *Towards Efficient Reinforcement Learning via Environment Measurement and Shaping*.
Advisors: Prof. Wenjie Li (Maggie), Prof. Wei Zhang. Joint doctoral training with the Eastern Institute of Technology (EIT), Ningbo.

Jinan University, Guangzhou, China — Sept 2020 to Jun 2024

B.Eng. in Cyberspace Security. GPA 88.6 / 100 · Rank 6 of 62 · Outstanding Graduate.

Publications

First-author

1. **Yanjun Chen**, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Exact Is Easier: Credit Assignment for Cooperative LLM Agents.** *arXiv:2603.06859*, 2026 (in submission).
2. **Yanjun Chen**, Yirong Sun, Xinghao Chen, Jian Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Integrating Chain-of-Thought for Multimodal Alignment: A Study on 3D Vision-Language Learning.** *arXiv:2503.06232*, 2025. [dataset]
3. **Yanjun Chen**, Xinming Zhang, Xianghui Wang, Zhiqiang Xu, Xiaoyu Shen, Wei Zhang. **Rethinking Soft Actor-Critic in High-Dimensional Action Spaces: The Cost of Ignoring Distribution Shift.** *arXiv:2410.16739*, 2024.
4. **Yanjun Chen**, Dawei Zhu, Yirong Sun, Xinghao Chen, Wei Zhang, Xiaoyu Shen. **The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models.** *EMNLP 2024*.

One further first-author manuscript under review, 2026.

Co-authored

1. Yunuo Liu, Dawei Zhu, Zena Al-Khalili, Dai Cheng, **Yanjun Chen**, Dietrich Klakow, Wei Zhang, Xiaoyu Shen. **PricingLogic: Evaluating LLMs Reasoning on Complex Tourism Pricing Tasks.** *EMNLP 2025*.
2. Xinghao Chen, Zhixin Sun, Wenjin Guo, Miao Zhang, **Yanjun Chen**, Yirong Sun, Hao Su, Yu Pan, et al. **Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning.** *Findings of ACL 2025*.
3. Yirong Sun, Dawei Zhu, **Yanjun Chen**, Eric Xiao, Xinghao Chen, Xiaoyu Shen. **Fine-Grained and Multi-Dimensional Metrics for Document-Level Machine Translation.** *NAACL 2025*.
4. Jun Xue, Junze Wang, Shanze Wang, Xinming Zhang, **Yanjun Chen**, Wei Zhang. **FastDSAC: Unlocking the Potential of Maximum Entropy RL in High-Dimensional Humanoid Control.** *arXiv:2603.12612*, 2026.
5. Yirong Sun, **Yanjun Chen**, Xin Qiu, Gang Zhang, Hongyu Chen, Daokuan Wu, Chengming Li, Min Yang, Dawei Zhu, Wei Zhang, Xiaoyu Shen. **SonicBench: Dissecting the Physical Perception Bottleneck in Large Audio Language Models.** *arXiv:2601.11039*, 2026.
6. Yirong Sun, Yizhong Geng, Peidong Wei, **Yanjun Chen**, Jinghan Yang, Rongfei Chen, Wei Zhang, Xiaoyu Shen. **LLaSO: A Foundational Framework for Reproducible Research in Large Language and Speech Model.** *arXiv:2508.15418*, 2025.
7. Xinghao Chen, Anhao Zhao, Heming Xia, Xuan Lu, Hanlin Wang, **Yanjun Chen**, Wei Zhang, Jian Wang, Wenjie Li, et al. **Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning.** *arXiv:2505.16782*, 2025.

8. Xianghui Wang, Xinming Zhang, **Yanjun Chen**, Xiaoyu Shen, Wei Zhang. **MA-ROESL: Motion-aware Rapid Reward Optimization for Efficient Robot Skill Learning from Single Videos.** *arXiv:2505.08367*, 2025.
9. Yuhan Hu, Yirong Sun, **Yanjun Chen**, Xinghao Chen, Xiaoyu Shen, Wei Zhang. **Breaking the Pre-Planning Barrier: Adaptive Real-Time Coordination of Heterogeneous UAVs.** *arXiv:2501.14488*, 2025.

Open Source

C3 github.com/Battam1111/C3

Reference implementation of *Exact Is Easier* (arXiv:2603.06859): exact per-decision credit via transcript replay, with a method-agnostic audit of credit quality.

AccuracyParadox-RLHF github.com/Battam1111/AccuracyParadox-RLHF

Official implementation of *The Accuracy Paradox in RLHF* (EMNLP 2024): reproducible training pipelines, evaluation harnesses, and reference reward models.

3D-CoT dataset huggingface.co/datasets/Battam/3D-CoT

Chain-of-thought annotations for 3D vision-language alignment, released with *Integrating Chain-of-Thought for Multimodal Alignment* (arXiv:2503.06232).

Teaching & Service

Teaching (TA)	COMP5221 Software Project Management · COMP1002 Computational Thinking and Problem Solving · COMP1411 Introduction to Computer Systems · COMP5567 Distributed Algorithms and Protocols for Blockchains (2024 to 2026).
Reviewing	ACL Rolling Review (ARR) 2026 · IEEE Transactions on Neural Networks and Learning Systems (TNNLS).

Technical Skills

Post-training	SFT and RLHF; PPO, DPO, GRPO; reward modeling and evaluation; RLVR; SAC and off-policy RL for continuous control.
Training infra	DeepSpeed (ZeRO), Megatron-LM; verl, OpenRLHF, Hugging Face TRL; LoRA / PEFT; multi-node multi-GPU training; vLLM inference.
Frameworks	PyTorch, Hugging Face Transformers / Datasets / Accelerate; Weights & Biases.
Programming	Python, C/C++, Bash; Git, Linux.
Languages	Mandarin (native), English (professional), Japanese (elementary).